

Problems Overview

Introduction

The Port of Dover is the UK’s busiest international ferry port, linking southeast England to France across a 34 km route [1]. It handles around 9 million passengers and over 2 million freight vehicles each year [2]. Its fast processing and frequent sailings, running roughly every 30 minutes, make it a critical cross-channel hub [3].

Impact

The port moves about £144 billion of goods annually, representing nearly one-third of UK–EU traded goods [4]. It also handles roughly 31% of all seaborne HGV traffic [4], making it essential for manufacturers, retailers, and logistics operators. Operational efficiency generates an estimated £3 billion in annual transport cost savings through faster crossings and more reliable schedules [4]. The port also supports thousands of jobs and underpins the regional economy in Kent [5].

Stakeholders

- **Dover Harbour Board:** Manages and operates the port [1].
- **Ferry Operators:** P&O Ferries, DFDS, and Irish Ferries [3].
- **Border Agencies:** UK Border Force and French Police Aux Frontières operate juxtaposed border checks [6].
- **Local and Transport Authorities:** Kent Police, National Highways, and the Kent Resilience Forum coordinate traffic and resilience planning [7].

Queuing Process

Arriving vehicles pass through check-in and passport control before entering marshalling lanes for boarding [1]. Traffic flow is monitored through real-time sensor systems [8]. During peak demand, external controls are used:

- **Dover TAP:** Holds HGVs on the A20 under a 40 mph limit to reduce town congestion [9].
- **Operation Brock:** Uses a section of the M20 as a controlled queuing zone during severe disruption [7].

Bottlenecks

Delays occur when demand exceeds processing capacity, often due to weather, ferry disruption, or extended post-Brexit passport checks. Cars may take around 90 seconds to clear French controls, and coaches up to 20 minutes [10]. This has resulted in severe peak delays, including six-hour queues for cars in 2022 and up to 18 hours for coaches in 2023 [11, 10]. Such congestion also disrupts local roads and affects nearby businesses [5].

Parameter	Meaning	Public Availability	Availability	Source / Method	Impact on Accuracy
$\lambda_v(t)$	Arrival rate for vehicle type v	Yes (aggregated)	(aggregated)	Dover Annual/Quarterly Statistics; DfT Counts	Critical
c_s	Number of servers/booths at stage s	Yes		Port layout diagrams	High
$\mu_{s,v}$	Service rate at stage s for type v	Partial (ranges)		RoRo academic studies; border processing reports	Very critical
$G_{s,v}$	Service-time distribution	No		Queueing assumptions (e.g., exponential/Gamma)	High
$p_{s \rightarrow s+1,v}$	Routing probability	Implicitly	Yes	Port process descriptions	Moderate
$p_{inspect,v}$	Inspection probability	No		Assumed from EU/UK risk-based rates	Very critical
$c_{a,s}$	Arrival variability	Partial		Derived from peak-to-average ratios	High
$c_{s,v}$	Service-time variability	No		Literature assumptions	Moderate
$H_{\max}$	Holding-area capacity	Yes (estimable)		Satellite imagery; port geometry	Moderate
L_s	Lane length	Yes		Satellite measurement	Low
β_s	Vehicle density (vehicles/meter)	Yes		Transport engineering	Low
B_f	Ferry capacity	Yes		DFDS/P&O vessel specifications	High
α_v	Lane-meters per vehicle type	Yes		Standard vehicle dimensions	Low
τ_f	Ferry departure times	Yes		Public ferry timetables	High
$x_{v,f}(t)$	Boarding decision variable	Model-defined		N/A	N/A

Table 1: Model parameters, public availability, and estimated impact on accuracy.

Mathematical Formulation

Sets

$$V = \{car, HGV, coach\}, \quad S = \{1, 2, 3\}$$

Arrival Process

Vehicle arrivals follow a nonhomogeneous Poisson process:

$$\lambda_v(t) = \text{arrival rate of type } v \text{ at time } t, \quad \lambda(t) = \sum_{v \in V} \lambda_v(t).$$

Service Process

For each stage s and vehicle type v ,

$$S_{s,v} \sim G_{s,v}(\mu_{s,v}),$$

where $1/\mu_{s,v}$ is the mean service time.

Number of servers:

$$c_s = \text{number of available booths at stage } s.$$

Traffic intensity:

$$\rho_s(t) = \frac{\lambda_s(t)}{c_s \mu_s}.$$

Routing

$$p_{s \rightarrow s+1,v} = \text{probability of advancing from stage } s \text{ to stage } s+1.$$

Inspection routing:

$$p_{2 \rightarrow 3',v} = p_{\text{inspect},v}, \quad p_{2 \rightarrow 3,v} = 1 - p_{\text{inspect},v}.$$

Queue Dynamics (G/G/c Approximation)

Effective arrival rate:

$$\lambda_s(t) = \sum_{v \in V} \lambda_v(t) \prod_{i=1}^{s-1} p_{i \rightarrow i+1,v}.$$

Expected waiting time using the Allen–Cunneen approximation:

$$W_{q,s}(t) \approx \frac{C_s \cdot \rho_s(t) \sqrt{2(c_s+1)-1}}{c_s \mu_s (1 - \rho_s(t))},$$

where

$$C_s = \frac{c_{a,s}^2 + c_{s,v}^2}{2}.$$

Expected queue length:

$$L_{q,s}(t) = \lambda_s(t) W_{q,s}(t).$$

Capacity Constraints

Holding-area capacity:

$$L_{q,3}(t) \leq H_{\max}.$$

Lane-length constraint:

$$L_{q,s}(t) \leq \beta_s L_s.$$

Ferry Boarding (Batch Process)

Let ferry f depart at time τ_f with capacity B_f .

$$\sum_{v \in V} \alpha_v x_{v,f}(t) \leq B_f.$$

Queue update after boarding:

$$L_{q,3}(t^+) = L_{q,3}(t^-) - \sum_{v \in V} x_{v,f}(t).$$

Example Objective: Minimise Total Waiting Time

$$\min \int_0^T \sum_{s \in S} L_{q,s}(t) dt.$$

References Cited

- [1] Port of Dover, “Port of dover overview.” <https://www.portofdover.com>, 2023. Accessed 2025.
- [2] UK Department for Transport, “Port traffic statistics 2023.” <https://www.gov.uk/government/statistics/port-freight-annual-statistics>, 2024. Passenger and freight volumes.
- [3] I. DFDS, P&O Ferries, “Dover–calais ferry services.” <https://www.dfds.com/en/ferry-routes/dover-calais>, 2024. Service frequency information.
- [4] Port of Dover, “Economic impact of the port of dover.” <https://www.portofdover.com/about/impact>, 2024. Trade, HGV share, and cost savings.
- [5] Kent County Council, “Economic significance of the port of dover to kent.” <https://www.kent.gov.uk/business/ports>, 2023. Employment and regional impact.
- [6] UK Home Office, “Uk–france juxtaposed border controls.” <https://www.gov.uk/government/publications/juxtaposed-controls>, 2023. Border process information.
- [7] Kent Resilience Forum, “Operation brock and kent resilience traffic plans.” <https://www.kentprepared.org.uk>, 2024. Traffic management and emergency planning.
- [8] Port of Dover, “Real-time traffic management at the port of dover.” <https://www.portofdover.com/operations>, 2022. Sensor and traffic flow systems.
- [9] National Highways, “Dover tap traffic management scheme.” <https://nationalhighways.co.uk>, 2023. A20 freight control system.
- [10] BBC News, “Easter travel delays at the port of dover.” <https://www.bbc.co.uk/news/uk-england-kent>, 2023. Coach and car processing times and delays.
- [11] BBC News, “Summer 2022 dover delays and critical incident.” <https://www.bbc.co.uk/news/uk>, 2022. Six-hour queues during peak travel periods.